

Cao Vu Hoang Long

☎ 0359697220 @ hoanglongcaovu@gmail.com  [Linkedin](#)  github.com/LongChuToa

EDUCATION

Ho Chi Minh City University of Technology, VNU-HCM

Ho Chi Minh City, Vietnam

Bachelor of Engineering in Computer Science

Aug. 2023 – Present

TECHNICAL EXPERIENCE

Software Engineering

Aug. 2025 – Dec. 2025

HCMC University of Technology, VNU-HCM

Ho Chi Minh City, Vietnam

- Modeled complex software systems utilizing extensive UML documentation, including use-case, sequence, class, and activity diagrams.
- Managed project phases utilizing Agile methodologies and Scrum frameworks
- Worked in a team to architect and prototype a university Tutor Support System, delivering UI mockups, sequence diagrams, and a working MVP.

Computer Networks

Aug. 2025 – Dec. 2025

HCMC University of Technology, VNU-HCM

Ho Chi Minh City, Vietnam

- Designed and implemented client-server communications as part of intensive, compulsory laboratory coursework.
- Co-developed a multi-threaded web server and reverse proxy from scratch utilizing pure Python sockets to implement TCP/IP and HTTP protocols directly.

Principles of Programming Language

Aug. 2025 – Dec. 2025

HCMC University of Technology, VNU-HCM

Ho Chi Minh City, Vietnam

- Understand mechanics of modern programming paradigms, managing memory environments, control flow, and data referencing.
- Engineered a complete compiler for "OPLang" a custom, statically-typed object-oriented programming language.
- Architected the backend logic for a custom programming language "OPLang", translating high-level OOP concepts into executable machine instructions.

PROJECTS

KongFuongChicken Website | *PHP, MySQL, Bootstrap, Docker, Composer, Git*

Feb. 2026 – May 2026

- Developed a full-stack F&B e-commerce website inspired by KFC, implementing user authentication, registration, session-based authorization, and database connectivity using PHP and MySQL.
- Designed and implemented the relational database schema, integrated UUIDv7-based user identification with `ramsey/uuid`, and optimized data handling using Prepared Statements and MVC architecture.
- Built news and admin management modules, including CRUD functionality for news content and administrative interfaces for managing website data and user interactions.
- Implemented input validation, secure authentication workflows, responsive UI components with Bootstrap, and deployment support using Docker and XAMPP environments.

File Sharing Web Backend | *Golang, Gin, PostgreSQL, Docker, nginx, Git*

Nov. 2025 – Jan. 2026

- Developed robust application security features, including complete user authentication and TOTP protocols.
- Deployed full-stack infrastructure using Docker containers for both the backend application and database.
- Built automated testing frameworks and CI/CD pipelines to accelerate reliable feature releases.
- Managed network traffic routing by provisioning a Linux virtual machine and configuring an NGINX reverse proxy.

YOLO:HOME Smart Home System | *Flutter, Flask, PostgreSQL, IoT, Adafruit IO*

Mar. 2026 – May 2026

- Developed the frontend interface for a smart home management system, enabling users to monitor environmental data and control IoT devices through responsive web/mobile dashboards built with Flutter.
- Collaborated with backend and IoT teams to integrate real-time sensor data, device status updates, and remote control functionalities through REST APIs and Adafruit IO cloud services.

TECHNICAL SKILLS

Languages: C/C++/C#, Python, SQL (MSSQL, Postgres, MySQL), JavaScript, HTML/CSS, PHP

Frameworks: React, Gin, Flutter

Developer Tools: Git, Docker, VS Code, Visual Studio, Eclipse, Figma, XAMPP, Unity